Invert Binary Tree

AlgoExpert · Medium · Binary Trees

The problem

- Mirror the tree by swapping every node's children.

Intuition

- Swap every node's children, recursively or with a queue.

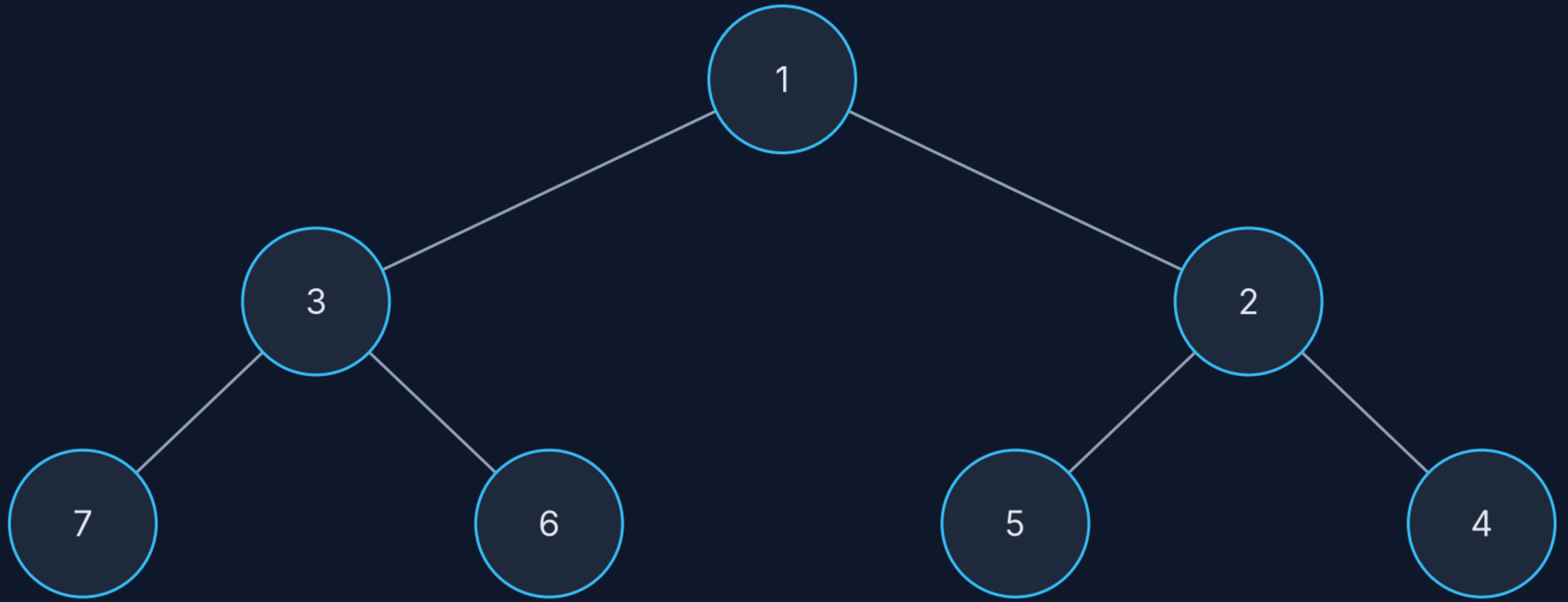
Brute force vs optimal

- Every node must be swapped once → O(n) is optimal
- Recursive (O(h) stack) or BFS queue (O(w) space)
- Order of visiting doesn't matter

Approach

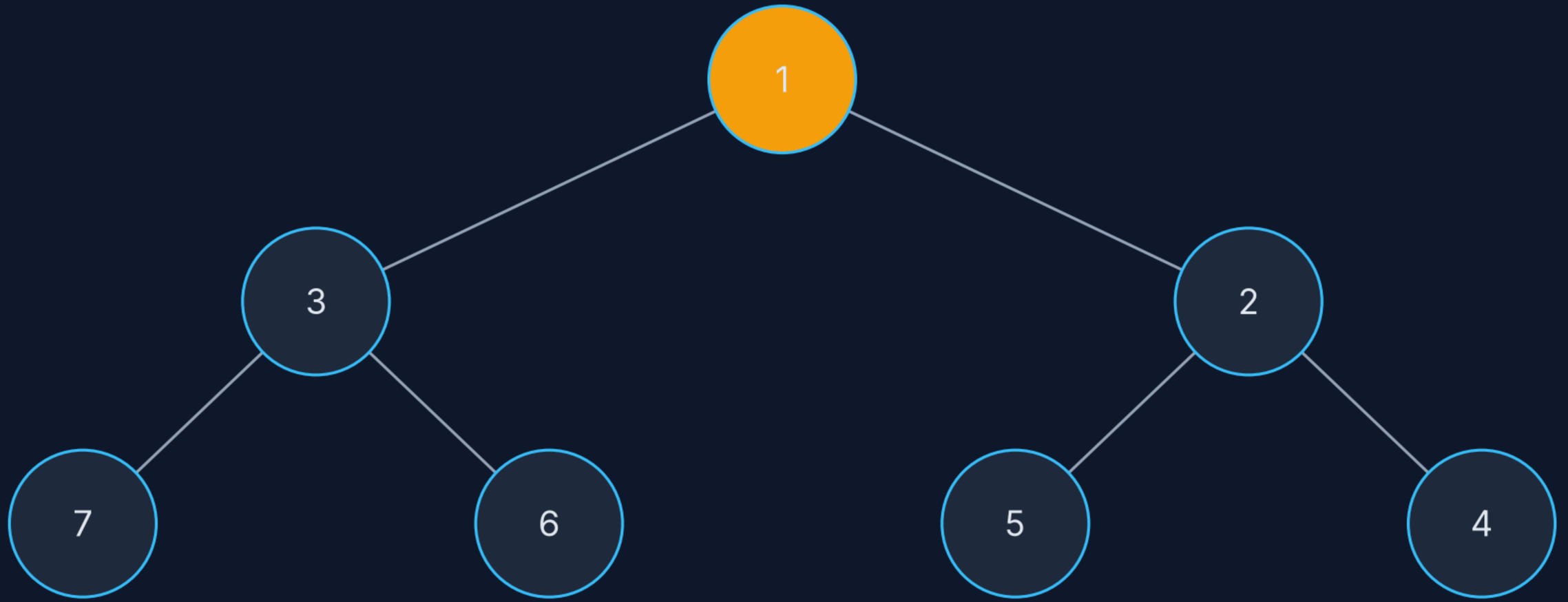
- 1. Visit each node
- 2. Swap left and right
- 3. Recurse into both children

Worked example



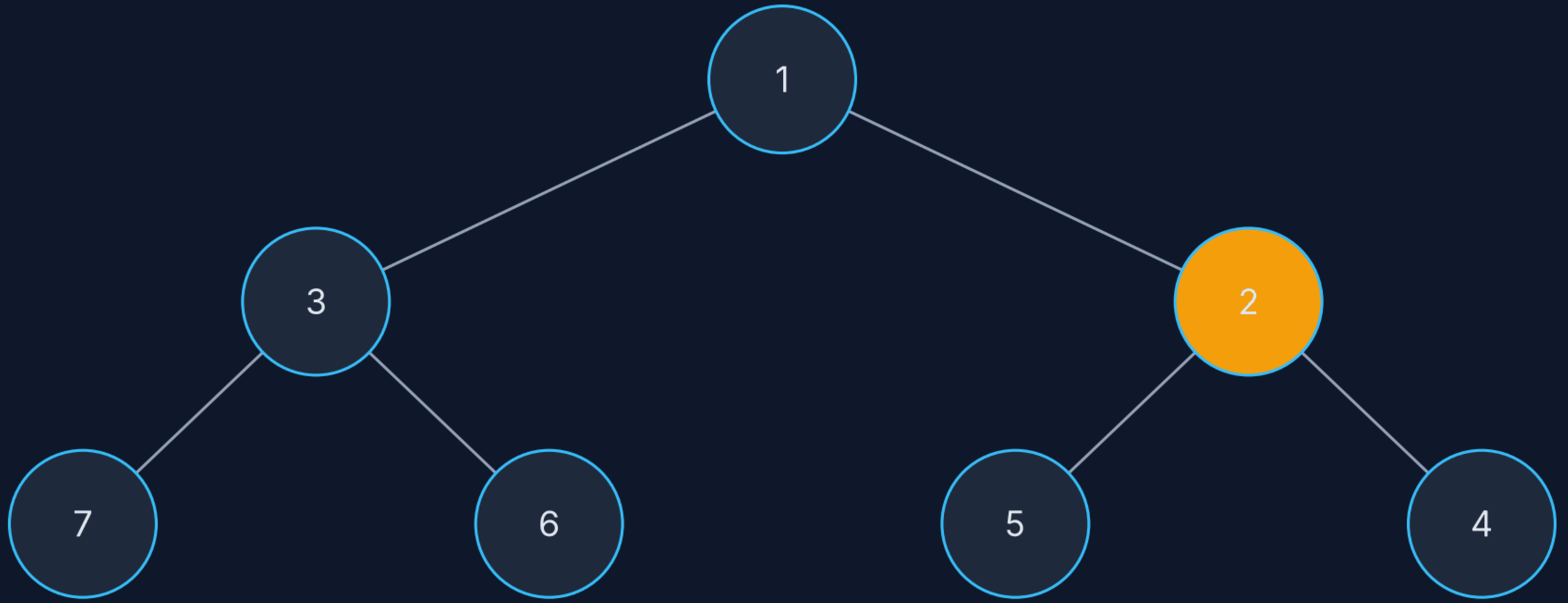
Original tree

Worked example



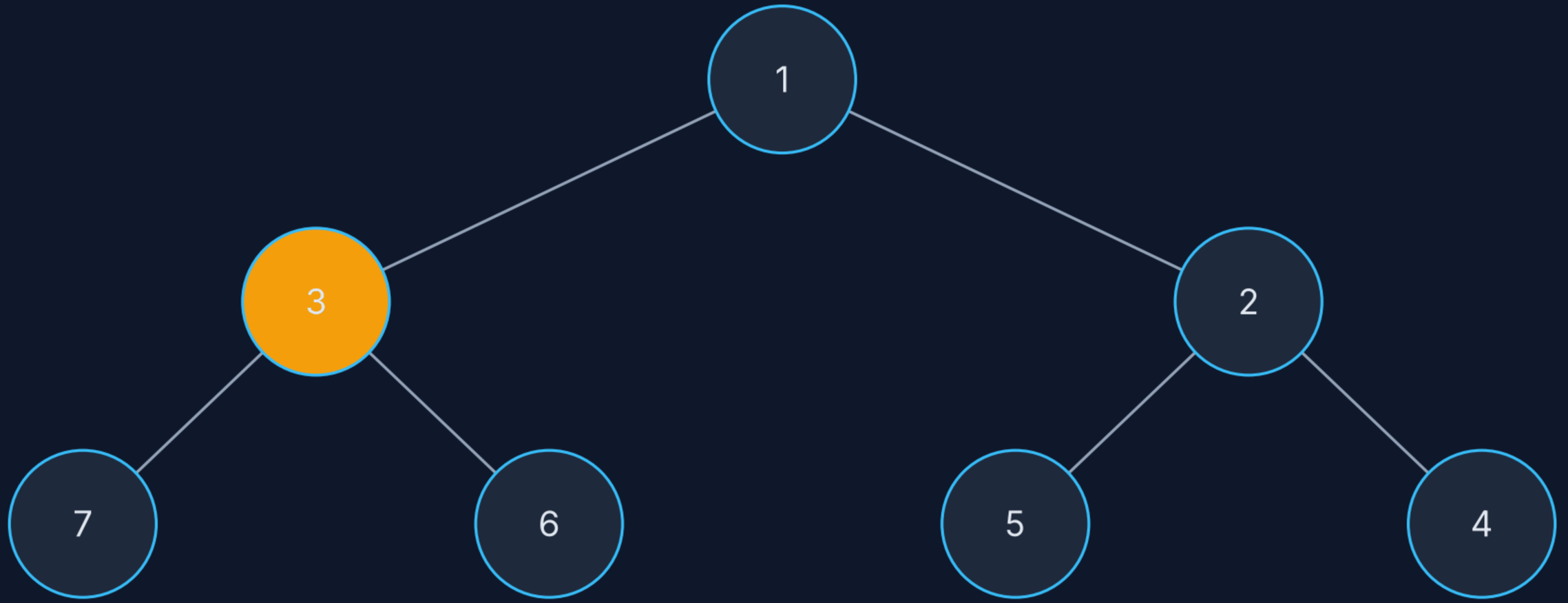
Swap children of 1

Worked example



Swap children of 2

Worked example



Swap children of 3

Java solution

```java
public class InvertBinaryTree {
    public static void invertBinaryTree(BinaryTree t) {
        if (t == null) return;
        BinaryTree tmp = t.left; t.left = t.right; t.right = tmp;
        invertBinaryTree(t.left);
        invertBinaryTree(t.right);
    }
}
```

Complexity

- Time: O(n)
- Space: O(h)

Edge cases & pitfalls

- Null children are swapped too
- Swap before or after recursion — both fine
- Deep trees: prefer iterative

Interview tips & follow-ups

- Mention the iterative BFS version
- Follow-up: check if two trees are mirrors (Symmetrical Tree)

Recap

- Swap every node's children, recursively or with a queue.
- Complexity: O(n)